

UF HEALTH CORE POLICY AND PROCEDURE

POLICY NUMBER: CP02.110

CATEGORY: Patient Care

REVIEW RESPONSIBILITY/DOCUMENT OWNER: Clinical Nurse Specialist

LAST REVIEWED: 2/27/2026

TITLE: Out-of-Hospital/Emergency Department (ED) Extracorporeal Cardiopulmonary Resuscitation (ECPR)

POLICY: Extracorporeal Cardiopulmonary Resuscitation (ECPR) has the potential to improve prognosis for a subset of out-of-hospital cardiac arrest patients during CPR and/or as part of CPR stabilization, including interventions to reverse the etiology of the arrest.

PURPOSE: This document outlines details of the Out-of-Hospital or Emergency Department (ED) Extracorporeal CPR (ECPR) program for a subset of patients in refractory Out-of-hospital cardiac arrest. It describes notification and alert criteria and roles & responsibility of those responding.

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DEFINITIONS:

- A. Extracorporeal CPR (ECPR) – the implantation of veno-arterial extracorporeal membrane oxygenation (VA-ECMO) in a patient who experienced a sudden and unexpected pulseless condition attributable to cessation of cardiac mechanical activity. ECPR is recognized in Extracorporeal Life Support Organization (ELSO) and American Heart Association (AHA) guidelines as a technique that can be considered in select patients in cardiac arrest.
- B. ERS – Emergency Resuscitation Service.
- C. Attending Physician: The physician primarily responsible for the patient's care.
- D. ED- Emergency Department
- E. Out-of-Hospital Cardiac Arrest (OHCA)

CORE PROCEDURES:

I. Determination of ECPR Candidate

- A. Alachua County Fire Rescue paramedics identify candidates in the field based on Medical Director-approved cardiac arrest protocol and contact the ERS Attending on-call directly. The ERS Attending on-call will notify the ED Coordinator with any patient accepted for ECPR evaluation.
 - 1. See [Appendix A](#) for OHCA process
- B. For patients not identified by EMS, the ED Attending Physician or ED RN will identify patients who meet inclusion criteria based on Medical Director-approved cardiac arrest protocol.
 - 1. Special consideration should be made for any cardiac arrest that does not respond to resuscitation within 5-10 minutes.
 - 2. Upon identifying a potential candidate, ED will call the ERS Attending on-call (352) 283-4198.
 - 3. If a patient does not meet criteria, the ERS Attending may still be called at the discretion of the ED Attending Physician for consideration of ECPR.
 - 4. The ERS Attending makes an independent decision to cannulate patients.
- C. The following individuals can place the call to the ERS Attending: Any ED APP, any ED resident physician, any ED attending physician, any ECMO perfusionist, and any RN. Attending to attending communication is preferred.
- D. After discussing the patient, the ERS Attending will decide on initiating the Out-of-Hospital ECPR protocol.

II. Response for ECPR Patient

- A. Communication across stakeholders

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1. The ERS Attending will call the ECMO Primer to bring supplies to bedside.
2. The ED Nursing Coordinator will facilitate communication to the nursing coordinators and 87 Charge RN to obtain an ICU bed for the ECPR patient.
3. The goal is to mobilize the resources to assign an ICU bed and to prepare for transport from the ED after patient is stabilized.
4. ECMO initiation will then proceed in the ED, with the intent that patients will proceed to the appropriate ICU once cannulated and stable for transfer.
5. If the patient is deemed to be a candidate for ECPR cannulation, cannulating physicians will agree on cannula configuration.
6. Patient will be admitted to the Emergency Resuscitation Service, under the ERS Attending on-call for the day of the cannulation.
7. Should it be deemed clinically appropriate, the ERS Attending will discuss the patient with the on call interventional cardiology attending for the consideration of coronary angiography after ECMO is initiated.
8. ECMO Coordinator or Primer is responsible for ensuring the ED ECPR ECMO Cart is properly stocked at all times, including restocking after each cannulation.

B. ICU Transfer

1. Patient will be transferred to an ICU bed as soon as possible.
2. A bed must be held open in Unit 87 ICU at all times for ECPR events. See CP02.111 In-hospital Extracorporeal Cardiopulmonary Resuscitation (ECPR).

III. Initiation of ECMO

- A. ECMO Capacity: The in-house ECMO primer will oversee the fluid nature of the organizations ECMO census and will be in ongoing communication with the ERS Attending and process stakeholders. See CP02.111 In-hospital Extracorporeal Cardiopulmonary Resuscitation (ECPR).
- B. Process for initiating ECMO: Upon notification by the ERS Attending, the ECMO primer will immediately proceed to the ED with a primed ECMO circuit and pump. ECMO primer will assist during the cannulation with all required materials and supplies from the ED ECPR ECMO cart. Once the ERS Attending has made the decision to cannulate, an ECPR Alert will be paged out.

IV. Quality control

- A. Case review will be conducted in the ECMO Quality meetings and reported through QPSR and CERC annually. In addition, the Director of the ERS directs an in-person review of each case with the faculty physicians of the ERS.

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ASSOCIATED POLICIES:

CP02.007 Clinical Emergency Response Plan

Applicable ECMO Policies

CP02.111 In-hospital Extracorporeal Cardiopulmonary Resuscitation (ECPR)

KEYWORDS:

CPR, ECPR, ECMO, Cannulation, ERS, AED, ED, Cardiac Arrest

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